Prototyping a RAG System for Digital Humanities: Ethical Considerations in AI Processing of Indigenous Data

Vieira, Miguel
jose.m.vieira@kcl.ac.uk
King's Digital Lab, King's College London, United Kingdom

Callaghan, Samantha
samantha.callaghan@kcl.ac.uk
King's Digital Lab, King's College London, United Kingdom

Ciula, Arianna
arianna.ciula@kcl.ac.uk
King's Digital Lab, King's College London, United Kingdom

Lu, Zihao
zihao.lu@kcl.ac.uk
King's Digital Lab, King's College London, United Kingdom

Ong, Tiffany
tiffany.ong@kcl.ac.uk
King's Digital Lab, King's College London, United Kingdom

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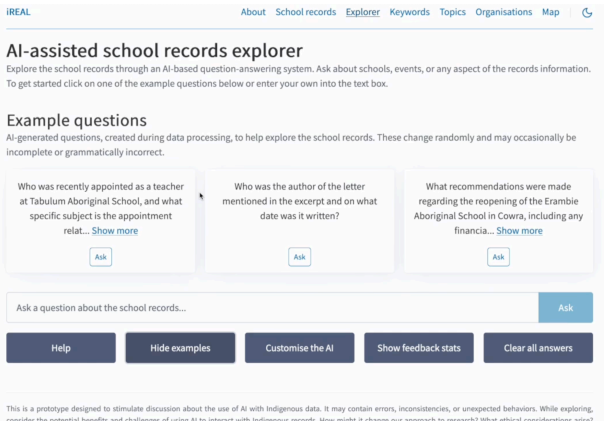
This poster presents work conducted by King's Digital Lab as part of iREAL (Indigenous Rights and Ethics in AI Libraries), a six-month scoping project funded by the Arts and Humanities Research Council's "Bridging Responsible AI Divides" scheme in the United Kingdom. iREAL aims to foster collaboration between Indigenous communities, librarians, and AI developers to ensure responsible inclusion of Indigenous knowledge in library AI systems, with a particular focus on Aboriginal and Torres Strait Islander communities in Australia.

Within this broader context, we developed a proof-of-concept Retrieval-Augmented Generation (RAG) system to explore the ethical implications of applying AI/ML technologies to Indigenous cultural data in Digital Humanities research. The prototype, developed as part of a dashboard, processes historical records from Aboriginal schools to stimulate critical discussion around AI ethics and cultural sensitivity in digital research methodologies.

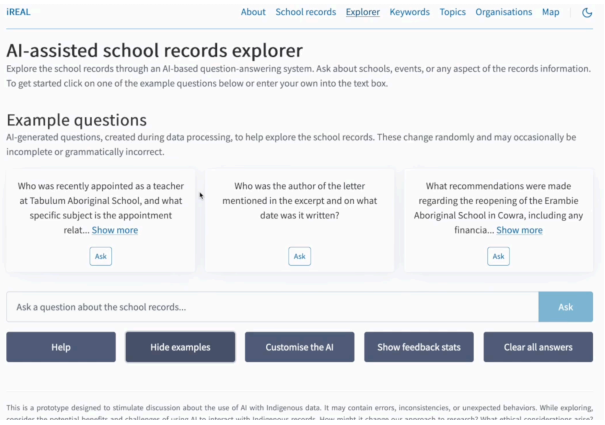
Our implementation combines large language models with a specialised knowledge base, incorporating key technical components including document processing pipelines, embedding and vectorisation systems, and multi-modal retrieval mechanisms. While exploring different language models for the prototype, we gained valuable insights into the trade-offs between model capabilities, computational requirements, and openness. This exploration highlighted how lighter, open-source models could make future implementations more sustainable and easier to maintain (Cordell, 2020). The system architecture emphasises modularity and configurability, allowing for iterative refinement based on stakeholder feedback. Development was informed by insights from a diverse project team and benefited from extensive feedback during two of the workshops, that contributed to the project's ethical framework and technical decisions.

From a technical perspective, the prototype was built using the LlamaIndex framework, implementing a hybrid search approach that combines vector similarity search with BM25 keyword matching for improved retrieval accuracy. The system uses the Mistral-7B language model for both text generation and embeddings, chosen for its balance of performance and resource requirements. Document processing uses a " semantic chunking " strategy to maintain context coherence, with the processed chunks stored in a Qdrant vector database. Observability is implemented via Arize Phoenix to provide insights into the RAG system's performance, improving the transparency and optimisation of the implementation.



RAG question/answer architecture diagram

The prototype's development involved a multidisciplinary team working over a short period, focusing on both technical implementation and ethical frameworks. Through this process, we identified several key considerations for future implementations, including the importance of balancing model capabilities with resource constraints, the need for continuous community involvement throughout development, the value of flexible architectures that can accommodate different types of cultural materials, and the importance of transparent documentation of AI system choices. The poster visualises key features including a user feedback system for response evaluation, integration with primary sources for verification, and transparent presentation of AI-generated insights. The system processes documents through a conversion pipeline from Word to JSON/Markdown formats, highlighting the technical challenges in standardising historical documents for machine processing.



Dashboard highlighting the RAG functionality

Significant challenges encountered during development included addressing semantic drift and maintaining cultural sensitivity while processing historical data. These challenges underscore the gap between basic RAG implementation and developing systems suitable for serious research applications. The project's findings emphasise the importance of community involvement in AI development processes, particularly when dealing with culturally sensitive materials (Lewis 2023).

While not intended for production use, this prototype serves as a valuable case study in applying AI/ML technologies to cultural heritage data. Through visual representations and interactive demonstrations at the poster session, we aim to stimulate discussion about ethical AI development, community involvement in technical projects, and the balance between innovation and cultural sensitivity in digital humanities research. The insights gained from this prototype will inform iREAL's broader goal of developing proposals for inclusive AI systems in libraries that respect Indigenous rights, knowledges, and data (Maiam nayri Wingara 2018).

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